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SEMANTIC-PRESERVING METRIC LEARNING FOR VIDEO-TEXT RETRIEVAL

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ABSTRACT

Video-text retrieval requires finding an optimal space for comparing the similarity of two different modalities. Most approaches adopt ranking loss as a primary training objective to find the space. The loss is only interested in bringing the samples annotated as pairs closer to each other without considering the semantic relevance of different samples. This rather causes even semantically similar pairs not to get close. To deal with the problem, we propose semantic-preserving metric learning. The proposed method entails the metric space where the similarity ratio between samples is proportional to semantic relevance between annotations. In the extensive experiments on video-text datasets, the proposed method presents a close alignment between the learned metric space and the semantic space. It also demonstrates state-of-the-art retrieval performance.

***Index Terms*—** Video-text retrieval, metric learning, semantic similarity

1. INTRODUCTION

Visual-semantic embedding is a method that maps vision and language into a common embedding space to understand both modalities. DeViSE [1] projected visual representation into a semantic embedding space. Subsequent, methods linked the two modalities in a common embedding space [2, 3, 4]. With significant advances of deep learning in natural language processing and computer vision, joint embeddings have been central to the tasks that require the understanding of two different modalities, such as image retrieval with natural language queries [2], captioning [5], and visual question-answering (QA) [6].

Here, we focus on joint embeddings for video-text retrieval. Various methods for video-text retrieval have been proposed [7, 8, 9]. Most of them share the same objectives, in which the difference between the similarity of the positive pair and the negative pair is larger than the margin. Such objectives entail mappings from different modalities into a joint embedding space, where corresponding inputs are closely mapped. However, learning the embedding space with a fixed margin for the limited ground truth does not reflect semantic relations and treats all negatives equally, which results in a discrepancy with human semantic relations, as shown in Fig. 1 (top right). This discrepancy is more severe in datasets with sparse ground truth pairs. To reduce semantic discrepancy, there are methods that weight/sample the training data [10, 11] or associate two modalities (*e.g.*, attention) to predict similarity [7, 8]. Nevertheless, because they formulate the relevance as a binary value, semantic relations among annotations is still not reflected in the training objectives.

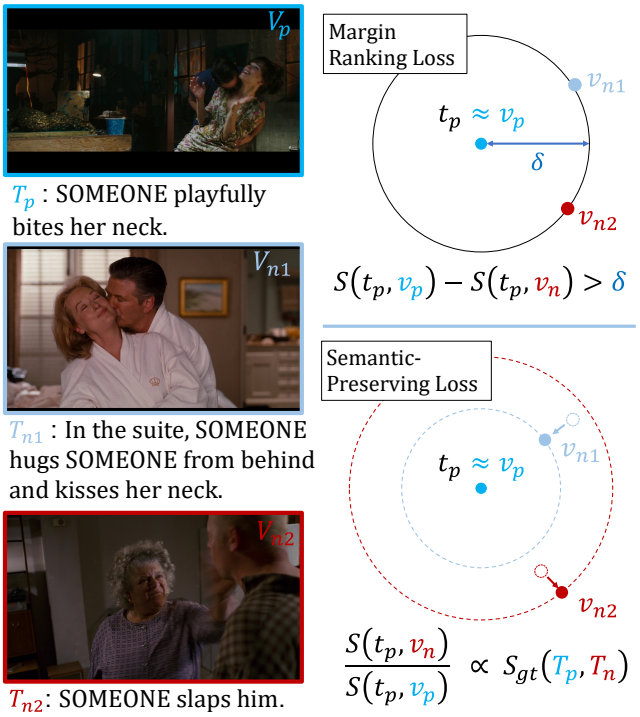

Fig. 1. Comparison of the rank-preserving method and the proposed method through sample pairs. Left: An example of a positive video-text pair V_p, T_p and two negative videos V_{n1}, V_{n2} with corresponding captions. Note that V_p and V_{n1} are semantically similar, but V_{n2} is not. Right: Conceptual diagram of the two methods in joint embedding space, where the small letters, v, t, and S mean the embedded corresponding modalities and similarity in the joint space. The ranking loss pushes negatives away from positives in the embedding space to a certain size while treating all negatives equally without considering semantic relevance. On the other hand, the proposed method is designed so that the ratio of similarities in the joint space is proportional to the similarity between the corresponding ground truth captions in the text space, S_{gt}, to reflect the relevance.

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In recent years, several methods have been proposed to enrich the label information by relaxing the binary relationship. Instead of learning the order with margin loss to increase precision indirectly, DeepRank [12] used a differentiable objective that could be learned directly. In [13], log-ratio loss was proposed to train the unimodal embeddings that follow semantic relations based on the pose of the person, caption, and segmentation labels of the images.

In the spirit of these methods, we propose training objectives for cross-modal embeddings to learn rich semantic information instead of learning only a binary relation of the corresponding pair. We reformulate the log-ratio loss [13] for cross-modal retrieval where each sample has only its own corresponding cross-modality as a positive. The learned embedding space is a metric space in which pairs with high similarity move closer to each other, and pairs with low similarity move farther away, as shown in Fig. 1 (bottom right). To show that the proposed method learns the embeddings preserving semantic relations as intended, we conduct extensive experiments. The contributions of this work are as follows:

- Instead of binary correspondence, we define a measure of semantic relations that allows us to obtain non-binary relations of labels.
- We propose the semantic-preserving metric learning, which regresses rich semantic relations even for cross-modal retrieval datasets with sparse ground truth pairs.
- Through extensive experiments, we show the closer alignment between the learned joint embedding space and the semantic space by the proposed method.

2. METHODOLOGY

2.1. Rank-Preserving Metric Learning

Consider a triplet (a, p, n), where a is an anchor sample, p is a positive sample, and n is a negative sample. Margin loss, which is the most common objective in metric learning, trains embeddings where p is closer to a than n. It is defined as

$$l_\delta(a, p, n) = max(0, D(f_a, f_p) - D(f_a, f_n) + \delta), \quad (1)$$

where D is a distance function for embeddings, δ is the margin of the loss, and f means embedding. Here, p is in the same class with a in unimodal retrieval. This loss aims at a *rank-preserving* metric, which maintains the positive-negative rank of triplets by learning embeddings to be spaced more than margins. As explained in Fig. 1, semantic discrepancy occurs when a fixed margin is applied to all triplets. To address this discrepancy directly, we propose a learning method in which semantic relation is preserved in the learned embedding space.

2.2. Semantic Relevance

Defining semantic relevance is necessary to be guided by semantic relations. Recently, language models pretrained on large amounts of unlabeled natural language data have shown performance close to humans in several downstream tasks. Predicting the similarity between sentences is one such downstream task. Among various ways to measure semantic similarity, we chose the angular similarity, $S_{gt} = 1 - \arccos(E(T_i) \cdot E(T_j))/\pi$, of two sentence embeddings $E(T_i)$ and $E(T_j)$ without fine-tuning. We use this simple method because the fine-tuning for each downstream dataset overfits and shows lower performance for out-of-domain datasets. This similarity measure does not reach the human level; however, semantic relations can be delivered even with this coarse similarity.

2.3. Semantic-Preserving Metric Learning

The proposed method is largely inspired by [13]. They proposed a log-ratio loss that regresses semantic labels for ground truth, instead of binary labels for unimodal image retrieval. The loss is calculated as

$$l_{lr}(a, p, n) = \left\{ \log \frac{D(f_a, f_p)}{D(f_a, f_n)} - \log \frac{D_{gt}(y_a, y_p)}{D_{gt}(y_a, y_n)} \right\}^2, \quad (2)$$

where D_{gt} is distance metric in the label space. The distance between two images in the learned metric space is proportional to their distance in the label space. Although the experiment for unimodal retrieval in [13] shows semantic-aware embedding space, it is limited in two ways. First, because there is no ground truth for the image-to-image retrieval dataset, the quantitative performance is evaluated based on the order of the semantic similarity, which is the same as that used in training. Second, distinguishing the gain by the log-ratio loss is difficult because the text annotations are also used compared with the ranking loss model.

We propose the semantic-preserving metric learning that trains *joint embedding* of the video-text retrieval dataset to follow the semantic space of annotations. In cross-modal retrieval, a recall performance can be measured based on ground truth pairs. Moreover, the text modality is fed equally to the ranking loss model. This clarifies the effect of the proposed method rather than experiments on unimodal retrieval. We adapt the previous method in two aspects to fit the cross-modal data. First, we modify the loss to be applicable to different modalities sharing the same semantic meaning. In the video-text retrieval dataset, positive for an anchor is fixed to only the corresponding cross-modal pair. Because the two different modalities share the semantic meaning, there is no semantic gap between the anchors and the positives. Therefore, the distance ratio cannot be regressed. To address this, the log-ratio loss is modified to use semantic relevance instead of distance. According to semantic relevance S_{gt}, the loss for the unimodal triplet in Eq. 2 is modified for the cross-modal triplet as

$$l_{t \to v}(p, n) = \left\{ \log \frac{S(t_p, v_n)}{S(t_p, v_p)} - \log S_{gt}(T_p, T_n) \right\}^2, \quad (3)$$

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Table 1. Comparison of rank-preserving metric learning (Rank) and the proposed semantic-preserving metric learning (Semantic) on six datasets. The gain is the amount of improvement using the proposed method, calculated by the sum of recalls.

Dataset	Loss	Text to video						Gain	Mean similarity of corresponding pairs
		R@1	R@5	R@10	MedR	MeanR	Sum of R		
LSMDC	Rank	10.6	25.5	33.1	33	122.8	69.2	+16.5%	0.14
	Semantic	12.7	29.2	38.7	25	103.2	80.6		0.23
DiDeMo	Rank	13.8	33.4	44.8	15	68.2	92.0	+11.6%	0.14
	Semantic	15.1	37.2	50.4	10	45.1	102.7		0.35
ANet	Rank	12.8	34.5	48.6	11	68.7	95.9	+9.2%	0.27
	Semantic	14.2	37.8	52.8	9	46.6	104.7		0.40
VTT	Rank	8.6	24.4	34.1	29	210.0	67.1	+7.6%	0.21
	Semantic	9.1	26.2	37.0	22	163.8	72.2		0.34
TGIF	Rank	5.9	14.8	20.6	122	729.1	41.3	+4.0%	0.20
	Semantic	6.0	15.4	21.6	93	563.9	43.0		0.36
VATEX	Rank	35.2	71.5	80.8	2	18.9	187.4	+2.0%	0.43
	Semantic	35.3	73.2	82.7	2	15.8	191.2		0.48

where S is an angular similarity of joint embeddings, v_p and v_n are positive and negative visual embeddings, T_p and T_n are positive and negative semantic annotations respectively, and t_p is the corresponding text embedding of the positive sample. This loss enables training of the cross-modal dataset. It also trains an embedding space in which the similarity ratio of the corresponding pair and the negative pair becomes close to the semantic similarity S_{gt}. Second, we reduce the side effects of non-ideal semantic metric by sampling the negatives for training. Ideally, the proposed method can learn the closest embedding space to the semantic space by training on all triplets without sampling. However, although the embedding learned without sampling maintains the semantic relation better, the retrieval performance declines. This is because our semantic measurement is not ideal, and outliers of annotations exist. With these experimental results and previous cross-modal studies [10], we sample the hardest negatives for training. It prevents overfitting to non-ideal metrics while improving retrieval performance. By applying bidirectional loss for cross-modal retrieval and hardest negative sampling, the overall loss for all samples is defined as

$$\mathcal{L}_{cross} = \sum_{p \in \mathcal{D}} \left\{ \max_{n \in \mathcal{B}_p} l_{v \to t}(p, n) + \max_{n \in \mathcal{B}_p} l_{t \to v}(p, n) \right\}, \quad (4)$$

where $\mathcal{D}$ is a set of sample indices in the dataset, $\mathcal{B}_p$ is a mini-batch of indices excluding p, and $l_{v \to t}$ is the loss of video query to text retrieval which is in opposite direction to $l_{t \to v}$.

3. EXPERIMENTS

Dataset. To evaluate the video-text embeddings of the proposed method, we perform experiments on six video datasets: MSR-VTT [14], TGIF [15], VATEX [16], LSMDC [17], DiDeMo [18], and ActivityNet Captions [19]. Because the sentences of ActivityNet Captions and DiDeMo are annotated for distinct moments of the videos, we evaluate the retrieval for the video-paragraph pairs as suggested in [20]. Following prior work, we report standard retrieval metrics including the recall at rank K (R@K; higher is better) and median, or mean rank (lower is better). Following the convention, we use K = 1, 5, and 10. We also calculate the sum of the recalls to reflect the overall retrieval performance.

Table 2. Distributions of features for each modality in the datasets. $N_{t:v}$ denotes the number of captions per video.

Dataset	$N_{t:v}$	$\bar{S}_v$	$\bar{S}_t$	$\bar{S}_{gt}$	D	Gain
LSMDC	1	0.73	0.82	0.58	1.77	+16.5%
DiDeMo	1	0.65	0.89	0.69	1.22	+11.6%
ANet	1	0.73	0.93	0.76	1.19	+9.2%
VTT	20	0.72	0.84	0.60	1.68	+7.6%
TGIF	1-3	0.70	0.90	0.66	1.45	+4.0%
VATEX	10	0.64	0.91	0.65	1.35	+2.0%

Implementation details. We use Dual Encoding [9] as the baseline among the existing architectures. The same hyper-parameters are used as in [9]. As for the video features, Kinetics pretrained I3D [21] features are used in VATEX, and ImageNet pretrained ResNet-152 [22] features are used in all other datasets. GloVe 300d word vectors [23] are used for the text feature. The pretrained *base-uncased* BERT model is used to measure the similarity, and special tokens are excluded. We use the hardest negative sampling in a mini-batch size of 128 in all experiments.

Effect of Semantic-Preserving Metric Learning. First, we compare the model trained with the proposed semantic-preserving metric learning and the baseline with the same structure but with ranking loss. The proposed method achieves consistent improvements across different datasets for all evaluation metrics in Table 1. Even in the datasets with a small improvement, the mean rank and median rank significantly decreased, indicating a better metric space than the baseline. Besides, the metric space that brings the two

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Table 3. Video-text retrieval comparison with state-of-the-art methods on the MSR-VTT test set. † denotes the method combining video-text modalities to measure similarity.

Method	Text to video				Video to text			
	R@1	R@5	R@10	MeanR	R@1	R@5	R@10	MeanR
VSE [1]	5.0	16.4	24.6	215.1	7.7	20.3	31.2	185.8
VSE++ [10]	5.7	17.1	24.8	300.8	10.2	25.4	35.1	228.1
Dual Encoding [9]	7.7	22.0	31.8	-	13.0	30.8	43.3	-
HGR [8]†	**9.2**	**26.2**	36.5	164.0	15.0	**36.7**	**48.8**	90.4
Ours	9.1	**26.2**	**37.0**	**163.8**	**15.2**	35.7	48.4	**80.8**

modalities closer has been learned through the result that the mean similarity of the corresponding pairs is increased. The normalized discounted cumulative gain (nDCG) [24], which evaluates the result by the semantic order, also shows improvement by the proposed method. We included the results of this nDCG evaluation in the supplemental material.

The effect of the proposed semantic-preserving learning appears differently depending on the dataset. To analyze this, we measure the distribution of the input video, text representations, and semantic annotations. It is measured by the cosine similarity of the features of each modality in randomly sampled training mini-batches of size B. The similarities between pairs, except corresponding pairs, are averaged. Formally, the mean intra-modal similarity of each modality is computed as

$$\bar{S}_f = \frac{1}{NB(B-1)} \sum_n \sum_{i \in B_n} \sum_{j \neq i} f_i \cdot f_j, \tag{5}$$

where B_n is the nth mini-batch of indices, N is the number of mini-batches, and $f \in \{v, t, gt\}$ are L2-normalized features of each modality. The video feature v is the average of frame features (*i.e.*, ResNet or I3D features), text feature t is the average of GloVe word embeddings of annotations, and semantic feature gt is BERT sentence embeddings of annotations. In Table 2, the performance improvement by the proposed semantic-preserving learning tends to be proportional as $\bar{S}_v$ and $\bar{S}_t$ increase, and $\bar{S}_{gt}$ decreases. This is because as the contrast of the semantic space increases, the effect of learning the non-binary labels of log-ratio loss becomes greater. The results also mean that, for inputs with similar scenes/words, semantic-preserving learning helps to distinguish them. To quantify this, we measure the initial discrepancy by $D = \bar{S}_v \bar{S}_t / \bar{S}_{gt}^2$ to be proportional to Eq. 3. The measured values are proportional to the degree of improvement. However, the improvement decreases for datasets with multiple captions per video. Because multiple captions allow for multiple video-text relations, various relations could be learned even with a fixed margin, which reduces the performance difference with the proposed method. However, initial discrepancy D is consistently proportional to the degree of improvement within datasets with multiple captions, which means that the proposed method solves the semantic discrepancy problem.

Comparison to State-of-the-Art Methods. We compare the proposed method to state-of-the-art (SOTA) methods. Table 3 compares the proposed method with SOTA methods on MSR-VTT. We only compared methods with a single video modality. There are methods of better performance with multiple modalities (*e.g.*, audio, scene text, and flow) of videos [25, 20, 26]. However, due to the differences in the amount of input information, we exclude them from the table because they cannot be compared on the same line. The proposed method consistently shows the SOTA level of performance not only in Table 3, but also in other datasets. Our method also efficiently retrieves the correspondences by the decoupled embeddings for each modality. The SOTA method which combines the two modalities to measure similarity improves performance but severely slows down the search speed. The supplemental material contains the comparisons with other datasets and an efficiency test.

4. CONCLUSION

In this work, we have introduced semantic-preserving metric learning. The proposed similarity measurement allows us to measure semantic relations between labels as continuous values to obtain rich information in semantic space. Unlike most video-text retrieval studies focused on structural improvement, the proposed method devised a training objective that learns a metric space closer to the semantic space. The proposed method enabled training with the limited correspondence of cross-modal retrieval and prevented overfitting to non-ideal metrics while improving retrieval performance. Extensive experiments showed a state-of-the-art retrieval performance and efficiency for large-scale video data. Our method can be applied not only to retrieval but also to any task that requires visual-semantic relations, *e.g.*, video QA and captioning, by simply changing the loss function. In future work, we plan to research a better metric for semantic space and learn visual representations involving human-level understanding.

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